

APURV KUSHWAHA

Robotics, Perception and Embedded Systems

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EDUCATION

Master of Science in Robotics 3.83/4.00 <i>University of Minnesota, Twin Cities</i>	Aug 2025 – May 2027 <i>Minneapolis, MN</i>
Bachelor of Technology in Engineering Physics <i>Indian Institute of Technology Guwahati</i>	Jul 2020 – May 2024 <i>Guwahati, India</i>

EXPERIENCE

Graduate Researcher <i>University of Minnesota, Twin Cities</i>	Apr 2026 – Present <i>Minneapolis, MN</i>
- Developing human-in-the-loop robot manipulation systems that combine Meta Quest 3 teleoperation with CAD/RGB-D tracking of a 3D-printed leader arm for a Trossen WidowX AI manipulator. - Implemented task-frame calibration, articulated-joint tracking, inverse kinematics, and safety-gated MuJoCo validation; validated Quest-based hardware control at 90 Hz with zero IK failures in a 51-second run.	
Lead Technical Engineer <i>Samaritan AI</i>	Nov 2024 – Jun 2025 <i>New Delhi, India</i>
- Led end-to-end development of a smart medical device, covering mechanical design, electronics integration, embedded firmware, and prototype validation. - Designed and fabricated mechanical parts using CAD and 3D printing, assembled electronic subsystems, and integrated sensors into a working prototype presented to 3+ investors and hospitals. - Developed ESP32 firmware for real-time sensing and device operation, and built 24-hour reliability tests that improved signal stability and reduced measurement inconsistency by 87%.	
Research & Development Engineer <i>Swachh.io</i>	May 2024 – Nov 2024 <i>New Delhi, India</i>
- Supported end-to-end development of environmental monitoring and pollution-control prototypes, contributing to design, testing, and system integration for real-world deployment. - Coordinated 4+ interns across mechanical, electronics, and software domains, and contributed simulation and validation work used in reports that helped secure a \$50K+ grant.	

SELECTED PROJECTS

Autonomous Car Parking Bot Code <i>ROS 2, Gazebo, Nav2, OpenCV</i>	Nov 2025
- Built an autonomous indoor parking system in ROS 2 and Gazebo, combining overhead-camera vehicle detection, Nav2-based navigation, and a towing robot with lift control. - Implemented coordinate transformation and fine-positioning logic for a parking lot, yielding highly precise orientation detection, ± 10 cm positioning accuracy, and 100% navigation success in tests.	
Warehouse Pick-and-Drop Optimization Code <i>A*, Hill Climbing, Simulated Annealing</i>	Dec 2025
- Developed a warehouse pick-and-drop optimization system by modeling task sequencing as an extended Traveling Salesman Problem with precedence constraints and A* path planning. - Implemented and compared greedy, hill climbing, and simulated annealing approaches, with experiments showing 5–12% route-cost improvement over the greedy baseline.	
Semantic-Aware Image Retargeting Code Demo <i>PyTorch, DINOv3, Depth Anything 3, OpenCV</i>	Dec 2025
- Built a semantic-aware image retargeting pipeline that combines saliency detection, depth estimation, and seam carving to resize images while preserving important content and visual composition. - Trained a DINOv3-based classifier on the PICD dataset (~43K images), achieving repo-reported 75–80% validation accuracy for composition-aware retargeting.	

SKILLS

Programming & Robotics: Python, C++, Embedded C, MATLAB, ROS 2, Gazebo, Path Planning, SLAM
Perception & AI: OpenCV, Sensor Fusion, Computer Vision, Machine Learning, Reinforcement Learning
Mechatronics: ESP32, Arduino, PCB Design, CAD, Embedded Systems, Sensor Integration, I2C, SPI, UART, CAN
Engineering Tools: Linux, Git, Docker, CMake, SolidWorks, Fusion 360, ANSYS, 3D Printing